Pablo Alonso Díaz Arancibia

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EDUCATION

Pontificia Universidad Católica de Chile **March 2020 – December 2025**
Bachelor's Degree in Astronomy *Santiago, Chile*

- **GPA: 6.02/7.0**; ranked among the top three students in my admission cohort.
- **Academic Honors:** graduated with two distinctions.

Certificate in Computing and Information Technology **December 2023**

- **Relevant coursework:** Advanced Programming, Databases, Software Engineering.

Bachelor of Science **December 2022**

RESEARCH EXPERIENCE

Pontificia Universidad Católica de Chile, Institute of Astrophysics **January 2025 – December 2025**
Research Assistant *Santiago, Chile*

- Analyzed spectroscopic data from 14 merging galaxies to study how galactic interactions influence central black hole activity and surrounding stellar populations.
- Built Python pipelines to process and analyze VLT/MUSE multidimensional spectral cubes, implementing statistical fitting models, numerical optimization, and Bayesian methods to characterize physical and kinematic properties and compare models.
  - **Stack:** Python (NumPy, SciPy, scikit-learn, Pandas, Astropy, lmfit, Matplotlib); pPXF; QFitsView; Git.

PROFESSIONAL EXPERIENCE

CONAF, Seed Center, Genetics and Entomology **January 2024 – February 2024**
Forest Inventory Manager *Chillán, Chile*

- Conducted and analyzed a forest inventory of over 1,000 specimens across 12 plots, recording dendrometric measurements and applying population-sampling methods to estimate mortality rates.
- Consolidated data from physical and digital sources in Excel and developed an SQL relational database prototype to organize the inventory, identifying and correcting errors and inconsistencies.
  - **Stack:** PostgreSQL, Visual Basic, Microsoft Excel.

PROJECTS

GALAH Galactic Archaeology

- Machine learning project built in Python with NumPy, Pandas, and scikit-learn. Used Random Forest models to classify galactic components, predict chemical abundances, and estimate stellar ages from GALAH DR4 data.

Binary Fraction in Star Clusters - Gaia DR3 + MIST

- Analyzed astronomical data from Gaia DR3 and MIST isochrones in Python to identify binary stars and estimate their fraction in globular clusters, studying how it relates to cluster age.

PKollector

- Full-stack web application built with React, Python (FastAPI), and PostgreSQL to manage Pokémon TCG collections. Includes card search and filtering, collection tracking, wish lists, and set-by-set progress statistics.

CERTIFICATIONS, SKILLS & LANGUAGES

- **Certifications:** Certificate in Computing and Information Technology (Pontificia Universidad Católica de Chile).
- **Skills:** Programming; Mathematical Modeling; Statistical Analysis; Automation.
- **Languages:** Spanish and English.